Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2001-II-Q02] | Part A (Q1-Q5)

All 2001 students study Spanish or French or both. Spanish enrollment is between 80% and 85%; French enrollment is between 30% and 40%. If the minimum and maximum possible overlap sizes are m, M, find $M - m$.

2001 名学生每人至少学习西班牙语或法语之一。西语人数在总数的 80% 至 85% 之间，法语人数在 30% 至 40% 之间。若双语人数最小、最大分别为 m, M，求 $M - m$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2003-I-Q03] | Part A (Q1-Q5)

For every two-element subset of $\{1, 2, 3, 5, 8, 13, 21, 34\}$, record its larger element. What is the sum of all recorded values?

对 $\{1, 2, 3, 5, 8, 13, 21, 34\}$ 的每个二元素子集记下较大元素，求所有记录之和。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2016-II-Q04] | Part A (Q1-Q5)

An integral $a \times b \times c$ box is assembled from unit cubes colored red, green or yellow. Each of its a layers perpendicular to the first dimension contains 9 red and 12 green cubes. Each of its b layers perpendicular to the second dimension contains 20 green and 25 yellow cubes. Find the least possible volume.

整数边长 $a \times b \times c$ 的长方体由红、绿、黄单位方块组成。垂直第一方向的每个切层均有 9 红、12 绿；垂直第二方向的每个切层均有 20 绿、25 黄。求最小可能体积。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2017-II-Q01] | Part A (Q1-Q5)

Count subsets of $\{1, \ldots, 8\}$ that are contained in neither $\{1, 2, 3, 4, 5\}$ nor $\{4, 5, 6, 7, 8\}$.

求 $\{1, \ldots, 8\}$ 的子集中，既不包含于 $\{1, 2, 3, 4, 5\}$ 也不包含于 $\{4, 5, 6, 7, 8\}$ 的个数。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2023-I-Q03] | Part A (Q1-Q5)

Forty nonparallel lines are drawn in the plane. Exactly $3, 4, 5, 6$ points have respectively $3, 4, 5, 6$ concurrent lines, and no point has more than six. How many intersection points have exactly two incident lines?

平面内有四十条两两不平行直线。恰有 $3, 4, 5, 6$ 个交点分别通过 $3, 4, 5, 6$ 条直线，无点通过超过六条。求恰通过两条直线的交点数。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2024-II-Q01] | Part A (Q1-Q5)

A town of 900 people has ownership totals $195, 367, 562, 900$ for four different items; everyone owns the fourth item. Exactly 437 people own two item types and 234 own three. How many own all four?

某镇 900 人对四类物品的拥有者总数依次为 $195, 367, 562, 900$，第四类人人都有。恰有 437 人拥有两类，234 人拥有三类。求四类全有的人数。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2014-II-Q09] | Part B (Q6-Q10)

Ten labeled chairs form a circle. Count subsets containing a run of at least three consecutive selected chairs.

十个固定座位围成圆，求含至少连续三个被选座位的子集数。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2018-I-Q09] | Part B (Q6-Q10)

Count four-element subsets of $\{1, \ldots, 20\}$ containing a distinct-element pair summing to sixteen and a distinct-element pair summing to twenty-four. The two pairs may overlap.

求一至二十的四元子集数，使其中有两个不同元素之和为十六，也有两个不同元素之和为二十四；两对可有公共元素。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2020-II-Q09] | Part B (Q6-Q10)

Six distinct people initially sit in the order $1, 2, 3, 4, 5, 6$. In how many new linear seating orders is every originally neighboring pair separated?

六人最初依次坐为 $1, 2, 3, 4, 5, 6$。重新排成一行时，原来相邻的任意两人都不得相邻，求排法数。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2022-I-Q06] | Part B (Q6-Q10)

How many integer pairs (a, b) make $3, 4, 5, a, b, 30, 40, 50$ strictly increasing while containing no four-term arithmetic progression, even as a nonconsecutive subsequence?

求整数对 (a, b) 的个数，使 $3, 4, 5, a, b, 30, 40, 50$ 严格递增，且任选四项均不能组成等差数列（不要求四项相邻）。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2010-II-Q11] | Part C (Q11-Q15)

Count the 3×3 zero-one matrices with exactly five ones in which at most one of the three rows, three columns, or two long diagonals is constant.

求恰含五个一、四个零的 3×3 矩阵数，使三行、三列、两条长对角线中至多一条上的三个元素全相等。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2022-I-Q12] | Part C (Q11-Q15)

For every ordered pair of equal-cardinality subsets $A, B \subseteq \{1, \ldots, n\}$, add $|A \cap B|$; call the sum S_n. Reduce S_{2022}/S_{2021} to p/q and find $(p+q) \bmod 1000$.

遍历 $\{1, \ldots, n\}$ 的所有等势子集有序对 (A, B)，将 $|A \cap B|$ 相加，所得为 S_n。将 S_{2022}/S_{2021} 约为 p/q，求 $p+q$ 除以 1000 的余数。

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Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

[AIME-2024-I-Q11] | Part C (Q11-Q15)

Independently color each vertex of a regular octagon blue or red, each with probability $1/2$. Find $m+n$ when the probability that some rotation sends every blue vertex to a formerly red position is m/n in lowest terms.

正八边形各顶点独立地以 $1/2$ 概率染蓝或红。若存在一次旋转使所有蓝点落在原先红点的位置，则称染色成功。成功概率为最简分数 m/n，求 $m+n$。

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